

Alireza Bayat Makou

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Research Interests

My research focuses on mechanistic interpretability of large language models. I study methods for discovering and evaluating circuits, including activation patching, causal interventions, and automated circuit-discovery approaches, with an emphasis on methodological reliability and the relationship between discovered structures and model behavior.

Education

Technical University of Darmstadt, Ph.D. in Computer Science, UKP Lab 2025 - Present

- **Research Focus:** Mechanistic interpretability and circuit discovery in LLMs.
- **Advisor:** Prof. Iryna Gurevych
- **Project Lead:** Murano, the mechanistic interpretability research design framework.
- **Mentoring:** Co-supervising Hanifi Akdag (M.Sc., TU Darmstadt), jointly with Jingcheng (Frank) Niu.

Leibniz University Hanover, M.Sc. in Computer Science (Track: AI) 2021 - 2025

- **Thesis:** SmartPEFT: Automatic Hyperparameter Tuning for Improved Parameter-Efficient Fine-Tuning of Large Language Models
- **First Examiner:** Prof. Marius Lindauer (AutoML Group, LUHAI)

Kharazmi University, B.Eng. in Computer Engineering (Track: Software Engineering) 2015 - 2021

- **Thesis:** Twitter Bot Detection via Large-Scale Account-Level Feature Analysis and Similarity-Based Clustering

Publications

Many Circuits, One Mechanism: Input Variation and Evaluation Granularity in Circuit Discovery.

Bayat Makou, A., Niu, J., Dutta, S., & Gurevych, I. *Transactions on Machine Learning Research* (TMLR), 2026. Featured Certification. [\[paper\]](#) [🔗](#)

Decoupled Agendas Under Repression: Social Media and State Media During COVID-19 in Iran.

Kermani, H., **Bayat Makou, A.**, & Behzadian Nejad, R. *Frontiers in Communication*, 10:1598405, 2025. [\[paper\]](#) [🔗](#)

LLM-based Rewriting of Inappropriate Argumentation using Reinforcement Learning from Machine Feedback.

Ziegenbein, T., Skitalinskaya, G., **Bayat Makou, A.**, & Wachsmuth, H. *ACL* 2024, pp. 4455–4476. [\[paper\]](#) [🔗](#)

Bots Versus Humans: Discursive Activism During the Pandemic in the Iranian Twittersphere.

Kermani, H., **Bayat Makou, A.**, Tafreshi, A., Mohamad Ghodsi, A., & Ataee, H. *Social Media + Society*, 9(4), 2023. [\[paper\]](#) [🔗](#)

Computational vs. Qualitative: Analyzing Different Approaches in Identifying Networked Frames During the Covid-19 Crisis.

Kermani, H., **Bayat Makou, A.**, Tafreshi, A., Ghodsi, A. M., Atashzar, A., & Nojoumi, A. *International Journal of Social Research Methodology*, 2023. [\[paper\]](#) [🔗](#)

Research Experience

Student Research Assistant May 2024 - Sep. 2024

L3S - Leibniz University Hanover

- Designed the system architecture and developed embedding-based retrieval components for a research search-engine project.
- Produced an internal literature review on text representations, embedding methods, and retrieval evaluation.

Student Research Assistant Apr. 2023 - Dec. 2023

Institute of Artificial Intelligence (LUHAI) - Leibniz University Hanover

- Contributed to an ACL 2024 project on LLM-based rewriting of inappropriate argumentation; supported text-style-transfer experiments, evaluation design, and analysis of rewritten social-media arguments.

Teaching Experience

Teaching Assistant, Introduction to Large Language Models

Winter Semester 2026/27 (assigned)

TU Darmstadt

- Will deliver lectures and exercises for the LLM portion of the course.

Natural Language Processing Instructor

Jun. 2022 - Oct. 2022

Filiger Academy of AI

- Designed and delivered “Natural Language Processing: A Practical Comprehensive Approach” with full curriculum and hands-on coding sessions (~30 hours, 50–60 students).

Natural Language Processing Instructor

Aug. 2020 - Mar. 2021

Loop Academy of AI

- Designed and delivered two NLP courses with full curriculum and hands-on coding sessions (~30 hours each, 50–60 students per course): “From One-hot to GPT-3: Introduction to Text Representation & Language Modeling”; “Fundamentals of Natural Language Processing”.

Industry Experience

Data Science / NLP Intern

Sep. 2022 - Mar. 2023

Vodafone

- Designed and developed a GPT-3-based summarization pipeline for chatbot-to-human handovers, achieving over 50% reduction in customer issue reading time.
- Collaborated on a Question-Answering / RAG component over internal wiki documents, reducing lookup time for support agents.
- Benchmarked GPT-family and encoder-decoder models (T5, BART) for dialogue summarization; co-designed a human-annotation interface for ongoing evaluation.

Machine Learning Engineer

Nov. 2020 - Jun. 2021

Astyage (Insurance Startup)

- Designed and implemented a data collection and annotation pipeline for constructing an insurance knowledge base and Question-Answering dataset.
- Built and deployed an insurance customer-service chatbot using Rasa with custom intent/entity fine-tuning and dialogue-story design; deployed at a major insurance provider.

Co-Founder & Machine Learning Engineer

Apr. 2019 - Sep. 2020

Gecko (NLP startup)

- Developed an opinion-mining and product-review summarization pipeline (sentiment aggregation, label extraction, explainable review-rejection) for a large e-commerce platform.
- Built a semantic-similarity and product-deduplication engine to flag duplicate listings created to manipulate search rankings, reducing manual moderation workload.

Technical Skills

Mechanistic Interpretability: circuit discovery; circuit evaluation; activation patching; causal interventions; feature probing; representation analysis; TransformerLens; NNsight.

NLP & Language Models: parameter-efficient fine-tuning; LoRA; adapters; text embeddings; dense retrieval; retrieval-augmented generation; dialogue summarization; model benchmarking; human evaluation.

AutoML & Optimization: Bayesian hyperparameter optimization; AutoML; active learning; Optuna.

Interpretable ML: SHAP; feature attribution; surrogate models.

Network Analysis & Graph ML: social network analysis; supervised and unsupervised bot detection; graph analysis; NetworkX, Gephi, PyTorch Geometric.

Programming & Infrastructure: Python; PyTorch; Hugging Face Transformers; scikit-learn; Git; Weights & Biases.

Languages

English: TOEFL (115/120) - C2

German: TestDaF (4/5) - C1